

The while Loop in C++

Computer Programming · UET NOWSHERA · Electrical Engineering

1 Objective

The objectives of this lab are:

- To understand the concept of the while loop in C++.
- To learn how a nested while loop works.
- To understand the do...while loop and its behavior.
- To write programs that execute repeatedly based on a condition.
- To apply loops in solving simple problems such as printing numbers and calculating sums.

2 Theoretical Background

The while Loop

A **while** loop is a control flow statement used in C++ to execute a block of code repeatedly as long as a given condition remains true. The condition is checked *before* executing the loop body. If the condition is true, the loop body executes and the process repeats until the condition becomes false.

Syntax

● ● ● C++ Code

```
while(condition)
{
    // statements
}
```

Working of the while Loop

- The condition is evaluated.
- If the condition is true, the loop body executes.
- After execution, the condition is checked again.
- The loop continues until the condition becomes false.

Nested while Loop

A nested while loop is a loop placed inside another loop. The outer loop controls how many times the inner loop runs, and the inner loop completes all its iterations for each iteration of the outer loop. Nested loops are useful for tables, patterns, or matrices.

●●● C++ Code

```
while(condition1)
{
    while(condition2)
    {
        // inner loop statements
    }
}
```

The do...while Loop

The **do...while** loop is similar to the while loop but with one important difference: the loop body is executed *at least once* because the condition is checked after the loop body executes.

Syntax

●●● C++ Code

```
do
{
    // statements
}
while(condition);
```

3 Software Used

IDE	DEV-C++ IDE
COMPILER	C++ Compiler (included in DEV-C++)

4 Experimental Tasks

Task 1 — Display Numbers Using while Loop

A C++ program was written using a while loop to display numbers from 1 to 10. The loop starts from 1 and continues until the value reaches 10. After printing each number, the counter is incremented by one. Each number is displayed on a separate line.

●●● C++ Code

```
#include <iostream>
using namespace std;

int main()
{
    int i = 1;
    while(i <= 10)
    {
        cout << i << endl;
        i++;
    }
    return 0;
}
```

Task 2 — Display Numbers Using do...while Loop

The same program was implemented using a do...while loop. The loop body executes first and the condition is checked afterward, guaranteeing the loop runs at least once.

●●● C++ Code

```
#include <iostream>
using namespace std;

int main()
{
    int i = 1;
    do
    {
        cout << i << endl;
        i++;
    }
    while(i <= 10);
    return 0;
}
```

Task 3 — Sum Using while Loop

A program was written that asks the user to enter a positive integer. A while loop calculates the sum of all numbers from 1 up to the entered number, then displays the total.

 C++ Code

```
#include <iostream>
using namespace std;

int main()
{
    int num, i = 1, sum = 0;
    cout << "Enter a positive integer: ";
    cin >> num;
    while(i <= num)
    {
        sum += i;
        i++;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

5 Conclusion

In this lab we learned how to use while loops and do...while loops in C++. We also learned how loops help repeat operations efficiently and solve problems like printing numbers and calculating sums. Nested loops were also covered as a technique for handling multi-dimensional problems such as tables and patterns.